

Mini E-Commerce Project (JDBC Based)

1. Use Case

Build a simple **E-Commerce Order System** where customers can browse products, add items to a cart, and place orders.

When an order is placed, the system should:

- Validate product stock availability
- Create a new order
- Update the product inventory
- Store the ordered items in the database

The system should use **JDBC Transactions** to ensure that all operations either complete successfully together or rollback if any step fails.

For database interaction:

- Use **PreparedStatement** for secure queries.
- Use **Batch Processing** for inserting multiple order items efficiently.

2. Functional Modules

Product Management

- View available products
- Display product details
- Show available stock quantity

Cart System

- Add products to the cart
- Remove products from the cart
- View items currently in the cart

Order Placement

- Create a new order for a customer
- Store order items in the database
- Validate stock before confirming the order

Inventory Update

- Reduce product stock after order confirmation
- Prevent orders if stock is insufficient

Order History

- View previous orders of a customer
- Display order details and items purchased

3. Low-Level Technical Requirements

- **Programming Language:** Java
- **Database Access:** JDBC
- **Database:** MySQL

Technical rules:

- Use **PreparedStatement** for all SQL queries (avoid using Statement).
- Use **JDBC Transactions** (commit / rollback) while placing orders.
- Use **Batch Processing** to insert multiple order items efficiently.

4. Database Tables

The system should include the following tables:

- **customers**
- **products**
- **orders**
- **order_items**
- **cart**